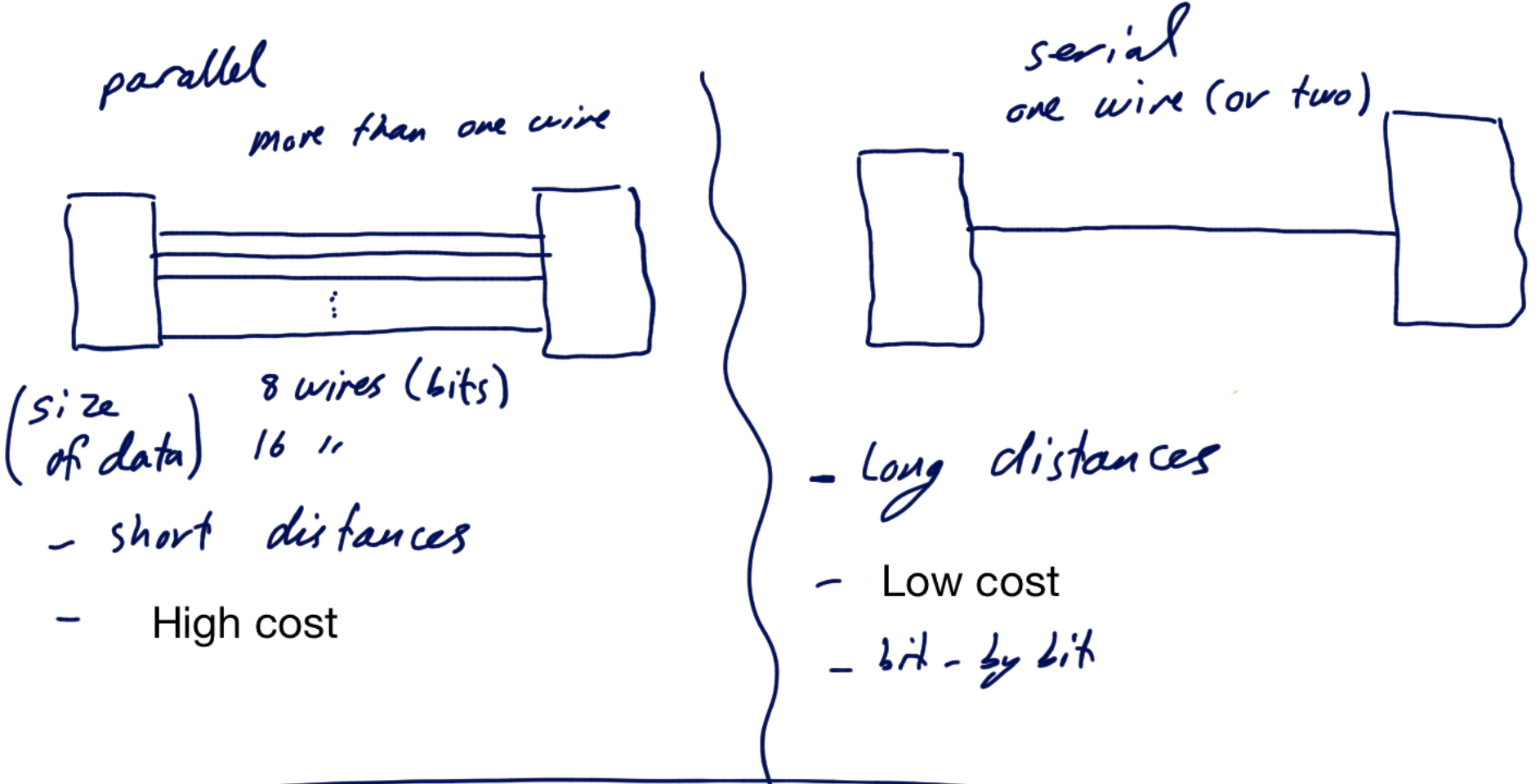
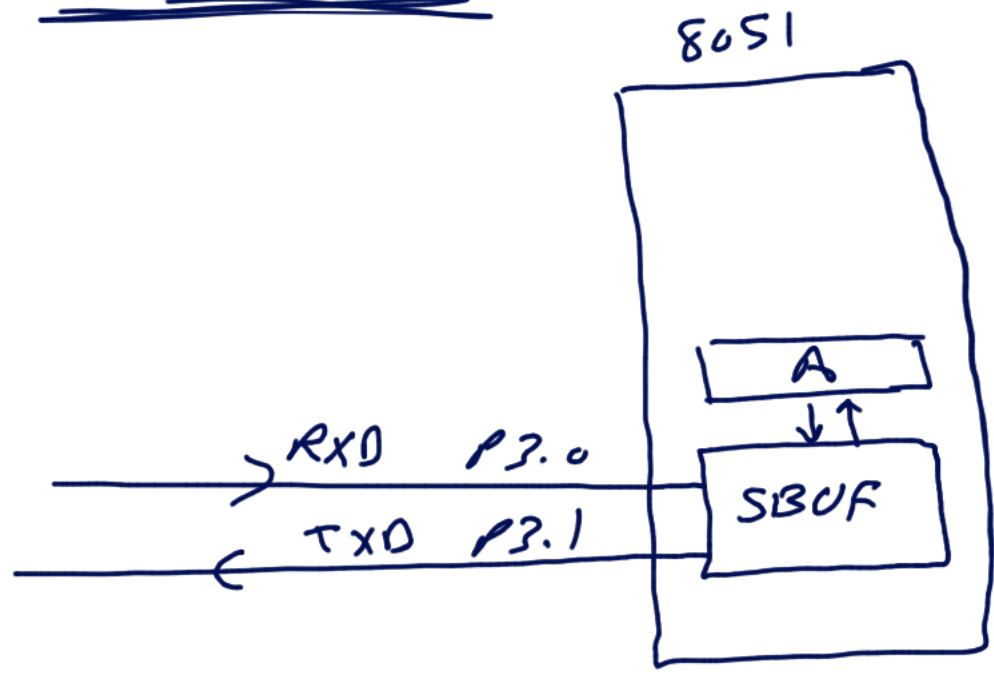


serial com

- compare between serial and parallel communication ?



serial com → serial port



- Modem:

- In serial communication: (Two types)

① Asynchronous: Transfer one byte (8 bit) at a time

② synchronous: Transfer a block of data at a time

- In any device $\rightarrow$ IC is used for serial com.

(a) UART: Universal Asynchronous Receiver Transmitter.

(b) USART: Universal Synchronous Asynchronous Receiver Transmitter.

- 8051 has UART $\leftarrow$

- In Asynchronous → protocol



- protocol is a set of rules that defines Asynchronous

com :

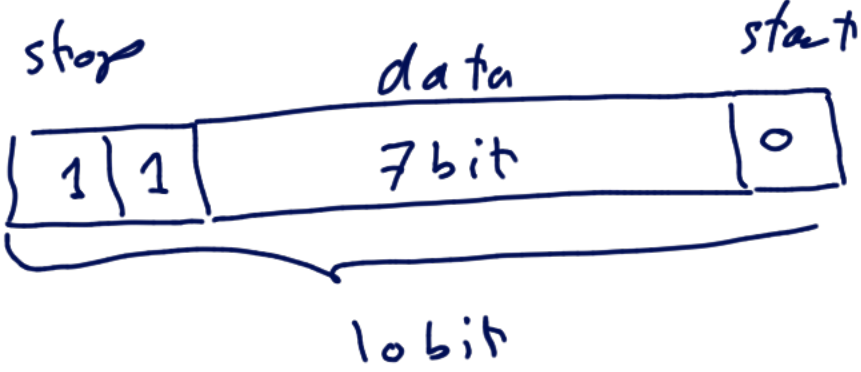
- ① when data begins
- ② size of data (8 bit)
- ③ when data ends
- ④ parity bit

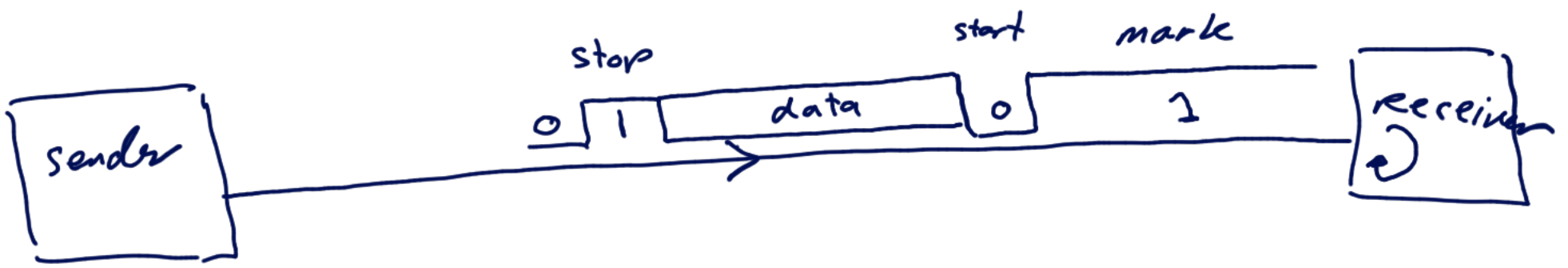
→ Framing : adding extra bits to the data.

new system



old system





parity bit : is used to detect errors

two types

- ① odd parity bit : # of 1's in data in addition to parity bit → odd
- ② Even parity bit : # of 1's in data in addition to parity bit → must be even

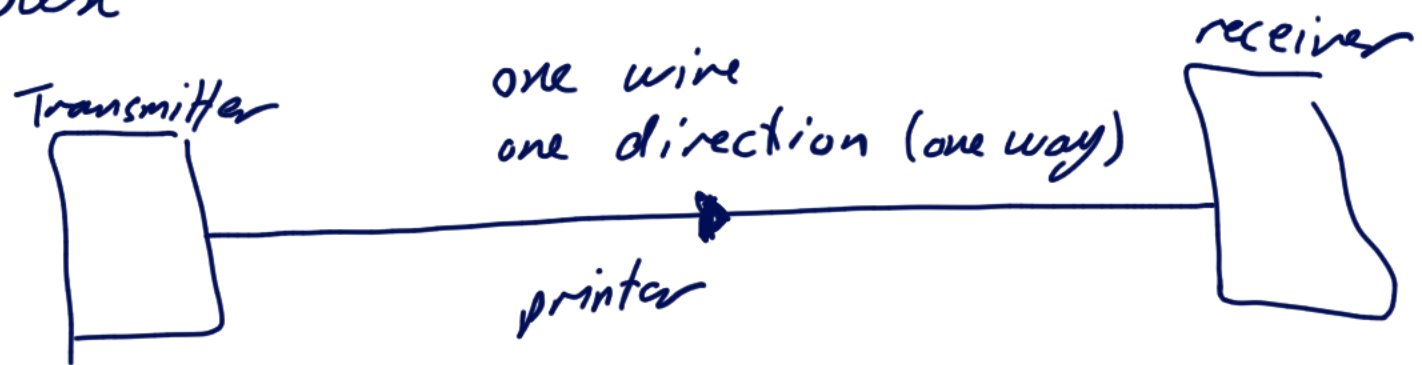
data
0 0 1 1 0 0 1 1

odd parity
1

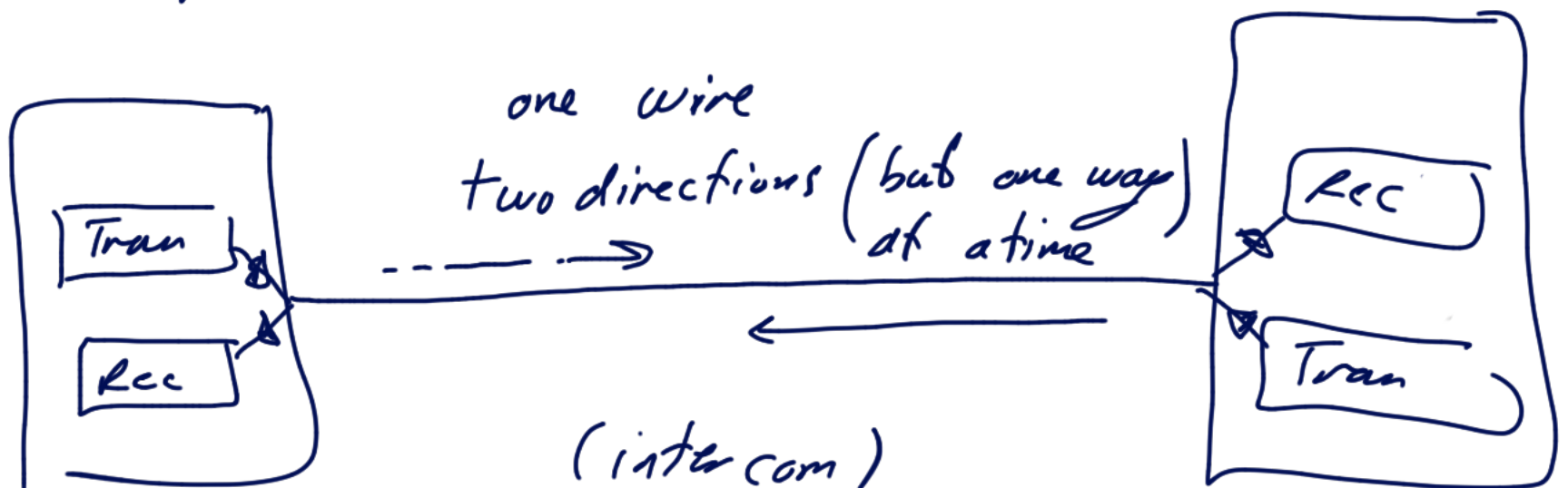
even parity
0

There are Three serial com modes:

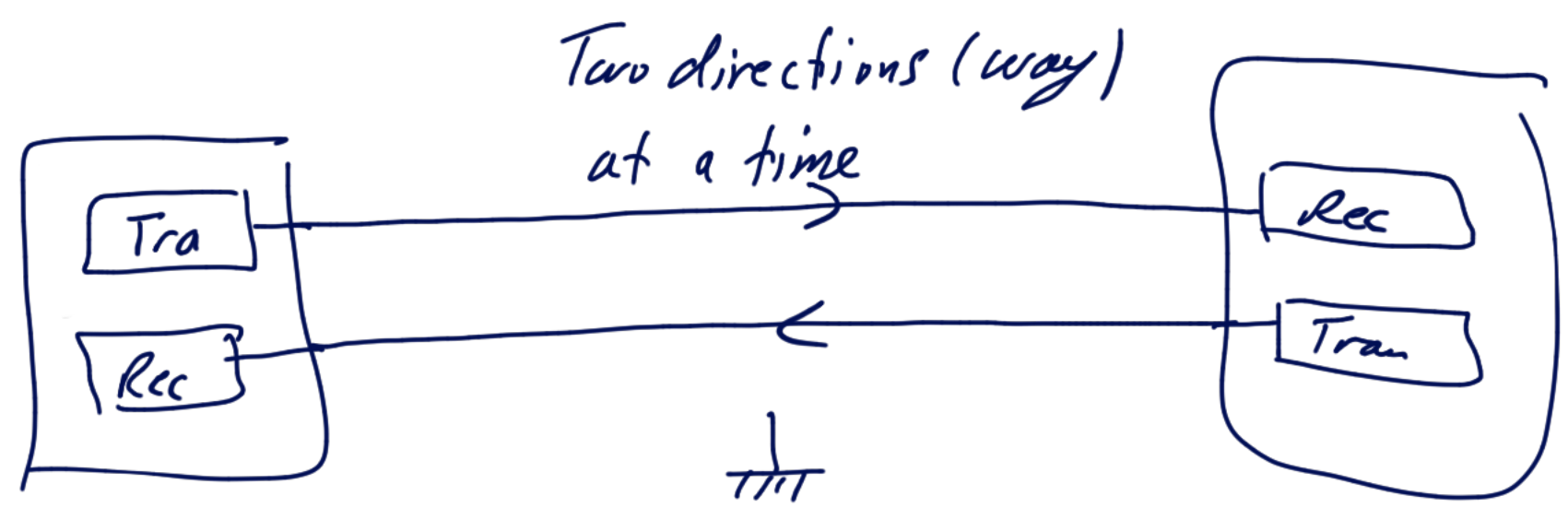
① simplex



② Half duplex :



③ Full duplex: Two wires



- RS232

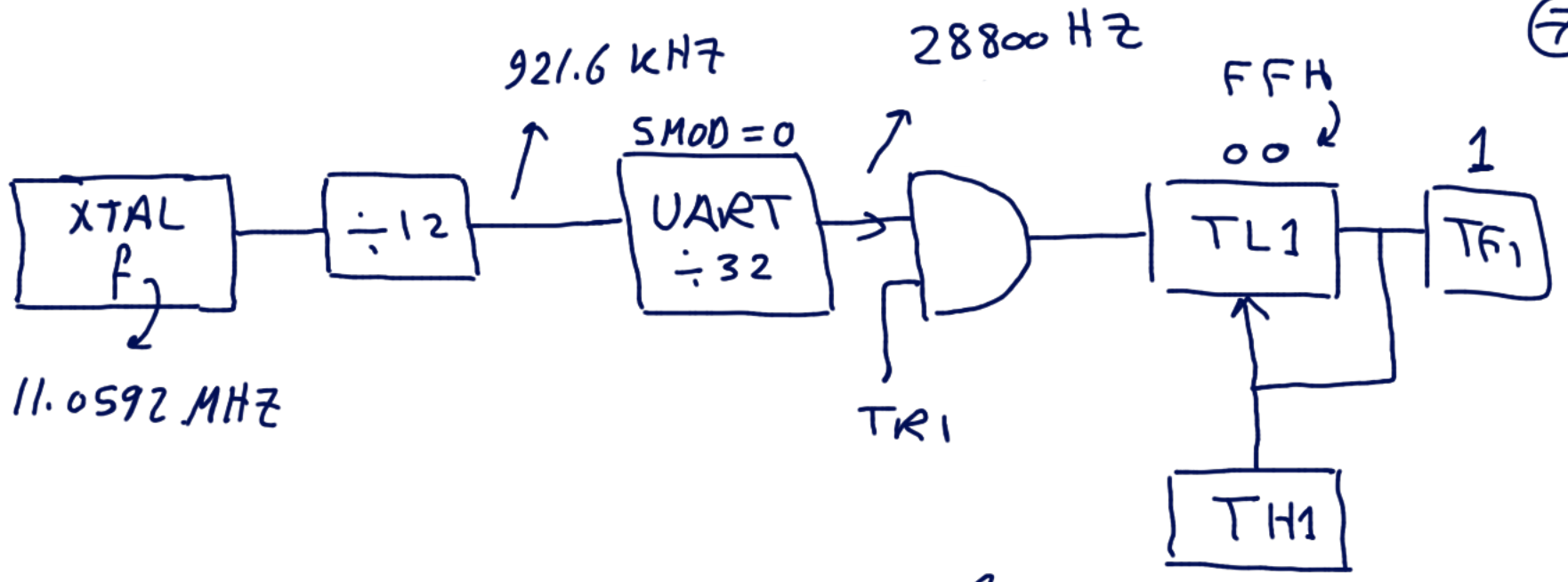
- MAX232:

- UART → in 8051

- band rate is programmable in 8051

↓
speed → bps $\frac{\text{bit}}{\text{sec}}$

- serial com → Timer 1 in mode 2

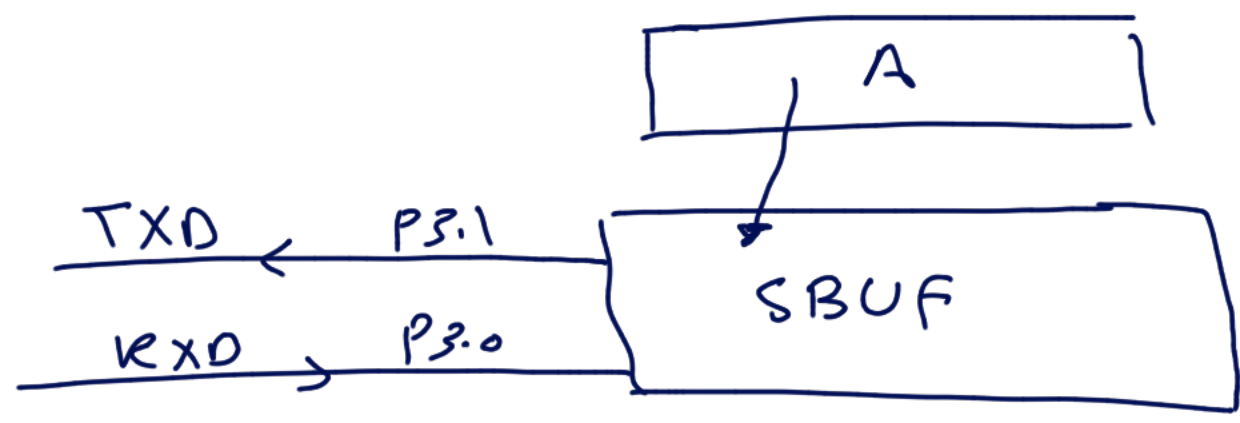


$$\text{UART Frequency} = \frac{\text{Machine Freq}}{32} = \frac{921.6}{32} = 28800 \text{ Hz}$$

$$\text{Timer 1 frequency} = 28800 \text{ Hz}$$

Assume XTAL = 12 MHz

$$\frac{12 \text{ MHz}}{12} = 1 \text{ MHz} \rightarrow \frac{1 \text{ MHz}}{32} = 31250 \text{ Hz}$$



if $TH1 = FFH \rightarrow \text{baud rate} = \frac{28800}{1} = 28800 \text{ bps}$
 $= -1$

if $TH1 = FDH \rightarrow \text{baud rate} = \frac{28800}{3} = 9600 \text{ bps}$
 $= -3$

if $TH1 = FAH \rightarrow \text{baud rate} = \frac{28800}{6} = 4800 \text{ bps}$
 $= -6$

SCON register: 8bit serial control

7	6	5	4	3	2	1	0
SM0	SM1	SM2	REN	/	/	TI	RI
0	1	0	1	0	0	0	0

SM0	SM1	serial mode
0	0	
0	1	1 → 8 data bit, one stop bit, start bit
1	0	
1	1	

serial mode 1

REN : Receive enable

REN = 0 $\rightarrow$ 8051 can only Transmit data (send)

REN = 1 $\rightarrow$ " " send and receive data.

TI: Transfer Interrupt flag $\rightarrow$ Transfer has been finished

RI: Receive Interrupt Flag $\rightarrow$ Receive " " "

MOV SCON, # 50H ; send/receive

MOV SCON, # 40H ; send only

Write a program to send msg. "Happy" serially at baud rate
9600 bps, continuously. Assume XTAL = 11.0592 MHz.

baud rate = 9600 bps

$$\frac{28800}{9600} = 3 \rightarrow TH1 = -3 \text{ or } FDH$$

ORG 0000H

MOV TMOD, # 20H

MOV TH1, # -3 ; 9600 bps

MOV SCON, # 50H

SETB TR1

NEXT: MOV DPTR, #List

BACK: CLR A

MOVC A, @A+DPTR

CJNE A, #0, CONT1

SJMP NEXT

CONT1: ACALL SEND

INC DPTR

SJMP BACK

Back: MOV A, # 'H'

ACALL SEND

MOV A, # 'a'

ACALL SEND

⋮
⋮

MOV A, # 'y'

ACALL SEND

SJMP BACK

SEND: MOV SBUF, A

HERE: JNB TI, HERE

CLR TI

RET

TI

1

List: DB "Happy", 0

END

Write a program to receive 12 byte of data serially (11)
at baud rate 4800 bps. Save received data in RAM
locations starting at 35H. XTAL = 11.0592MHz.

$$\frac{28800}{4800} = 6 \rightarrow TH1 = -6 \text{ or FAH}$$

```
ORG 0000H
MOV TMOD, #20H
MOV TH1, #-6
MOV SCON, #50H
SETB TR1
MOV R2, #12
MOV R0, #35H
```

```
BACK: ACALL RECV
      MOV @R0, A
      INC R0
      DJNZ R2, BACK
      SJMP $
```

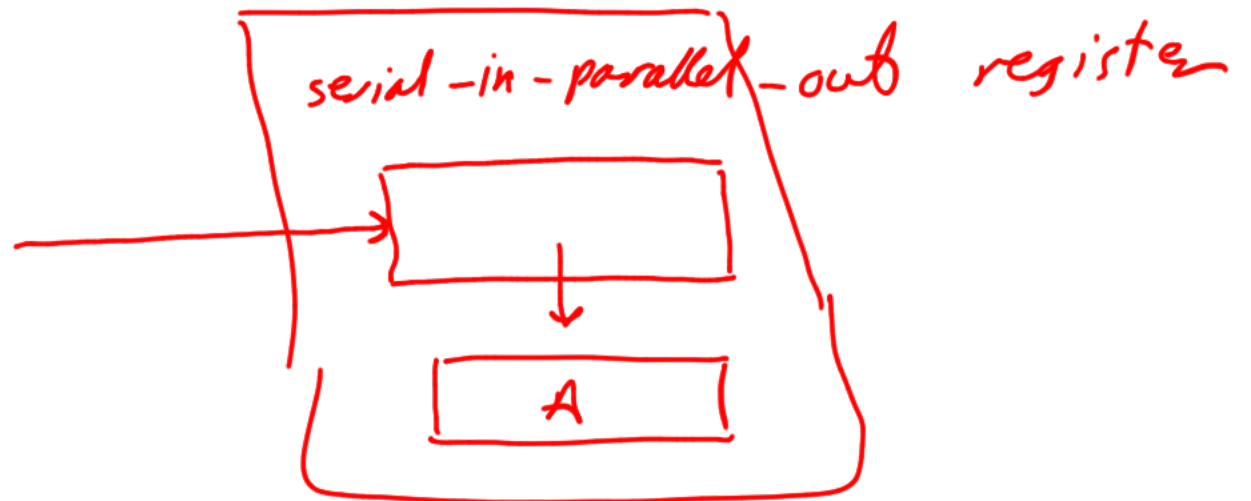


```
RECV: JNB RI, RECV
      MOV A, SBUF
      CLR RI
      RET
      END
```

KT [2]

Doubling baud rate

Receiver



Transmitter

